PAPER ID-310865

Subject Code: BEC055

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
ELECTRONICS SWITCHING

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	Define reed electronic switching system	1	K1
b.	Define trunking in switching systems.	1	K1
c.	What is a crosspoint in a space switch?	2	K2
d.	State the difference between GoS and Quality of Service (QoS).	3	K3
e.	Mention any two functions of inter-register signaling.	4	K1
f.	What is CCITT?	4	K1
g.	What is end-to-end flow control	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Describe Incoming Traffic and Service Time Characterization in detail	1	K2
b.	Explain the concept of Time Division Switching. Describe the working of a Time Slot Interchanger (TSI) with suitable diagrams.	2	K2
c.	Discuss the Erlang-C formula and explain how it is used to estimate delays in switching systems.	3	K3
d.	Describe the principle of Stored-Program Control (SPC). Explain centralized and distributed SPC with their advantages.	4	K2
e.	Explain in detail Routing Control and Flow Control.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Differentiate between circuit switching and packet switching.	1	K3
b.	Explain the term switching system. With the help of diagram explain various elements of switching system.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Differentiate between three stage STS and TST switching. Determine the implementation complexity a 2048-channnel STS switch implemented for 16 TDM Links with 128 channels on each link. The desired maximum blocking probability is 0.002 for channel occupancies of 0.1.	2	K3
b.	Describe the concept of automatic switching system. Also explain space division and time division switching.	2	K2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Derive the equation of Grade of service and blocking probability of lost call cleared service (LCC).	3	K3
b.	Derive Birth death process with neat sketch.	3	K3

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6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain the principles of Common Channel Signaling (CCS). Discuss its advantages over channel-associated signaling.	4	K2
b.	Describe the concept of reliability and availability conditions of processors in telecommunication exchange. A processor has MTBF = 2000 hrs and MTTR = 4 hrs. Calculate:(a) Availability(b) Unavailability	4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Describe statistical multiplexing in packet-switched networks. Compare it with time-division multiplexing.	5	K2,K3
b.	Write a short note on: 1. Banyan Network Switch 2. Memory-Space Memory switch	5	K2

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